

Homework #4

Math 104: Intro to Analysis

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0 Various Theorems

0.1 Limit Proof I

For $p, q, \alpha, C \in \mathbb{R}$ where $\alpha > 1$ and $C > 0$, $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$.

Proof. Let $\varepsilon > 0$. Let $\delta = \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}}$, which is greater than zero because $\alpha - 1 > 0$ and $C > 0$. Suppose that $0 < |p - q| < \delta$. Substitute in our value of δ to find $0 < |p - q| < \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}}$. Now, raise all sides to the positive power $\alpha - 1$, then multiply by C .

$$0 < |p - q| < \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}} \quad (1)$$

$$0 < |p - q|^{\alpha-1} < \frac{\varepsilon}{C} \quad (2)$$

$$0 < C |p - q|^{\alpha-1} < \varepsilon \quad (3)$$

Since $0 < C |p - q|^{\alpha-1}$, $C |p - q|^{\alpha-1} = \left| C |p - q|^{\alpha-1} \right|$. Therefore, $\left| C |p - q|^{\alpha-1} \right| < \varepsilon$. Ergo, $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$. OEΔ

0.2 Limit Proof II

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0 \quad (4)$$

Proof. Let $\varepsilon > 0$. Let $\delta = \frac{1}{\varepsilon} > 0$. Suppose that $x > \delta$. Multiply both sides by the reciprocals of x and δ to find that $\frac{1}{\delta} > \frac{1}{x}$. Substitute in the value of δ to find that $\frac{1}{\frac{1}{\varepsilon}} = \varepsilon > \frac{1}{x}$. Since $\delta > 0$ and $x > \delta$, $x > 0$, meaning $\frac{1}{x} > 0$, so $\frac{1}{x} = \left| \frac{1}{x} \right|$. So $\varepsilon > \left| \frac{1}{x} \right|$. Therefore, $x > \delta$ implies $\varepsilon > \left| \frac{1}{x} \right|$. Ergo, $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. OEΔ

0.3 Alternate Derivative Definition

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable.

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (5)$$

Proof. Recall the definition of $\frac{df(x)}{dx}$.

$$\frac{df(x)}{dx} = \lim_{c \rightarrow x} \frac{f(c) - f(x)}{c - x} \quad (6)$$

This means that for all $\varepsilon > 0$, there exists $\delta > 0$ such that $|c - x| < \delta$ implies $|f(c) - f(x)| < \varepsilon$. Let $h = c - x$. Solving for c , $c = x + h$. We can therefore conclude that $|h| < \delta$ implies $|f(x+h) - f(x)| < \varepsilon$. This fits a definition of a new limit.

$$\frac{df(x)}{dx} = \lim_{c \rightarrow x} \frac{f(c) - f(x)}{c - x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (7)$$

This completes the proof. OEΔ

0.4 Alternate Mean Value Theorem

Let f be a real, continuous function on $[x, x+h]$ which is differentiable in $(x, x+h)$, then there is a point $b \in (x, x+h)$ at which

$$f(x+h) - f(x) = hf'(b). \quad (8)$$

Proof. Recall the Mean Value Theorem. If f is a real continuous function on $[x, y]$ which is differentiable in (x, y) , then there is a point $b \in (x, y)$ at which

$$f(y) - f(x) = (y-x)f'(b) \quad (9)$$

Let $h = y - x$. By extension, $y = x + h$. Therefore, for $b \in (x, x+h)$,

$$f(x+h) - f(x) = hf'(b) \quad (10)$$

OEΔ

Let f be a real, continuous function on $[x-h, x]$ which is differentiable in $(x-h, x)$, then there is a point $c \in (x-h, x)$ at which

$$f(x) - f(x-h) = hf'(c). \quad (11)$$

Proof. This is very similar to the last. Recall the Mean Value Theorem. If f is a real continuous function on $[y, x]$ which is differentiable in (y, x) , then there is a point $c \in (y, x)$ at which

$$f(x) - f(y) = (x-y)f'(c) \quad (12)$$

Let $h = x - y$. By extension, $y = x - h$. Therefore, for $c \in (x-h, x)$,

$$f(x) - f(x-h) = hf'(c) \quad (13)$$

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Note that these are respectively equivalent to:

$$f'(b) = \frac{f(x+h) - f(x)}{h} \quad (14)$$

$$f'(c) = \frac{f(x) - f(x-h)}{h} \quad (15)$$

0.5 Mean in the Middle

For $a, b \in \mathbb{R}$, if $a < b$, then $a < \frac{a+b}{2} < b$.

Proof. Let $a, b \in \mathbb{R}$ where $a < b$. Add b to both sides and find $a+b < 2b$. Divide both sides by two to find $\frac{a+b}{2} < b$.

Take $a < b$. Add a to both sides and find $2a < a+b$. Divide both sides by two to find $a < \frac{a+b}{2}$.

Ergo, by the transitive property, $a < \frac{a+b}{2} < b$. OEΔ

1 Problem 1

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : (X, d_X) \rightarrow (Y, d_Y)$ is called *Lipschitz continuous* if there is a constant $L > 0$ so that

$$d_Y(f(p), f(q)) \leq Ld_X(p, q)$$

for all $p, q \in X$. For $\alpha > 0$, we say f is *Hölder continuous (with exponent α)* if there is a constant $C > 0$ so that

$$d_Y(f(p), f(q)) \leq Cd_X(p, q)^\alpha$$

for all $p, q \in X$. In particular, Hölder continuous with exponent 1 means the same thing as Lipschitz.

- (a) Fix a point $p \in X$. Show that the map $g : X \rightarrow \mathbb{R}$ defined by $g(x) = d(x, p)$ is Lipschitz continuous, with constant $L = 1$.
- (b) Prove that any Hölder continuous function (with any exponent $\alpha > 0$) is uniformly continuous.
- (c) Prove that if f is Lipschitz continuous and bounded, then f is Hölder continuous with exponent α , for any $0 < \alpha \leq 1$.
- (d) Let $[a, b]$ be a closed interval, and suppose $f : [a, b] \rightarrow \mathbb{R}$ is C^1 (meaning its first derivative exists and is continuous on $[a, b]$). Show that f is Lipschitz continuous.
- (e) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is Hölder continuous with exponent $\alpha > 1$. Show that f is constant.

1.1 Solution (a)

Proof. Since the codomain of g is $\mathbb{R}$, we can use the standard metric as d_Y . From the triangle inequality, we know that $d_X(x, y) + d_X(p, x) \geq d_X(p, y)$ and $d_X(x, y) + d_X(p, y) \geq d_X(p, x)$. We can reorder both of these to find that $d_X(x, y) \geq d_X(p, y) - d_X(p, x)$ and $d_X(x, y) \geq d_X(p, x) - d_X(p, y)$. Consolidating these, we can apply the absolute value to find that $d_X(x, y) \geq |d_X(p, y) - d_X(p, x)|$. This is also known as the inverse triangle inequality. Since the absolute value is the usual metric over $\mathbb{R}$, this means that $d_X(x, y) \geq d_{\mathbb{R}}(d_X(p, y), d_X(p, x))$. Since $g(x) = d_X(p, x)$, $d_X(x, y) \geq d_{\mathbb{R}}(g(y), g(x))$. Therefore, g is Lipschitz continuous with constant $L = 1$. OEΔ

1.2 Solution (b)

Proof. Let f be Hölder continuous with exponent α for $\alpha > 0$.

$$d_Y(f(p), f(q)) \leq Cd_X(p, q)^\alpha \tag{16}$$

Let $\varepsilon > 0$ and let $\delta = \sqrt[\alpha]{\frac{\varepsilon}{C}}$, which allows that $\delta > 0$ since $C > 0$. Suppose that for some $p, q \in X$, $d_X(p, q) < \delta$. We can substitute in our value of δ and isolate ε .

$$d_X(p, q) < \delta = \sqrt[\alpha]{\frac{\varepsilon}{C}} \tag{17}$$

$$d_X(p, q)^\alpha < \frac{\varepsilon}{C} \tag{18}$$

$$Cd_X(p, q)^\alpha < \varepsilon \tag{19}$$

Since $d_Y(f(p), f(q)) \leq Cd_X(p, q)^\alpha$, this means $d_Y(f(p), f(q)) < \varepsilon$. Therefore, $d_X(p, q) < \delta$ implies $d_Y(f(p), f(q)) < \varepsilon$. Ergo, f is uniformly continuous. OEΔ

1.3 Solution (c)

Proof. Let $0 < \alpha \leq 1$, and let f be Lipschitz continuous. Let $p, q \in X$.

$$d_Y(f(p), f(q)) \leq L d_X(p, q) \quad (20)$$

Since f is bounded, then there exists a number $M \in \mathbb{R}$ and a point $f(q) \in Y$ for some $q \in X$ such that for all $p \in X$, $d_Y(f(p), f(q)) < M$. Suppose we take three cases: $d_X(p, q) = 1$, $d_X(p, q) > 1$, and $d_X(p, q) < 1$.

Case 1: $d_X(p, q) = 1$. Anything real to the power of 1 is just 1, so if we set $C = M$, then it will always be true that $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$.

Case 2: $d_X(p, q) < 1$. If $d_X(p, q) < 1$, then for $0 < \alpha \leq 1$, $d_X(p, q) \leq d_X(p, q)^\alpha$. Therefore, if we set $C = L$, then $L d_X(p, q) \leq L d_X(p, q)^\alpha$, and therefore $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$.

Case 3: $d_X(p, q) > 1$. If $d_X(p, q) > 1$, then for all $0 < \alpha \leq 1$, $d_X(p, q)^\alpha > 1$. Therefore, if we set $C = M$, then $C d_X(p, q)^\alpha > M$, and by extension it will always be true that $d_Y(f(p), f(q)) < C d_X(p, q)^\alpha$.

Therefore, depending on the value of our inserts, we can take $C = \max\{M, L\}$ as a constant and f will be Hölder continuous. OEΔ

1.4 Solution (d)

Proof. $[a, b]$ is a subset of the real numbers, so we will be using the usual metric. If f is C^1 , then its first derivative exists and is continuous on $[a, b]$. Intervals are compact in the real numbers, so $[a, b]$ is compact. Since continuous functions on compact sets are bounded, $f'([a, b])$ is bounded. There exists $M \in \mathbb{R}$ such that for all $x \in [a, b]$, $|f'(x)| < M$. Without loss of generality, suppose $p \leq q$. By the mean value theorem, there exists $c \in (p, q)$ so that $f'(c) = \frac{f(q) - f(p)}{q - p}$. Since $c \in [a, b]$, $|f'(x)| < M$, so $\left| \frac{f(q) - f(p)}{q - p} \right| < M$. By the limit laws, $\frac{|f(q) - f(p)|}{|q - p|} < M$. Multiply both sides by $|q - p|$ to find that $|f(q) - f(p)| < M |q - p|$. Since $\mathbb{R}$ uses the usual metric, this means for any $x, p \in X$, $d_{\mathbb{R}}(f(p), f(q)) \leq L d_{[a, b]}(p, q)$ for any $p, q \in [a, b]$. Therefore, f is Lipschitz continuous. OEΔ

1.5 Solution (e)

Proof. Let f be Hölder continuous with exponent $\alpha > 1$. There exists a constant C such that for all $p, q \in \mathbb{R}$:

$$d_{\mathbb{R}}(f(p), f(q)) \leq C d_{\mathbb{R}}(p, q)^\alpha \quad (21)$$

Note that $\mathbb{R}$ uses the usual metric.

$$|f(p) - f(q)| \leq C |p - q|^\alpha \quad (22)$$

Divide both sides by $|p - q|$.

$$\frac{|f(p) - f(q)|}{|p - q|} = \left| \frac{f(p) - f(q)}{p - q} \right| \leq C |p - q|^{\alpha-1} \quad (23)$$

By the definition of the absolute value, $\left| \frac{f(p) - f(q)}{p - q} \right| \geq 0$.

$$0 \leq \left| \frac{f(p) - f(q)}{p - q} \right| \leq C |p - q|^{\alpha-1} \quad (24)$$

Take the limit of all sides as q approaches p .

$$\lim_{q \rightarrow p} 0 \leq \lim_{q \rightarrow p} \left| \frac{f(p) - f(q)}{p - q} \right| \leq \lim_{q \rightarrow p} C |p - q|^{\alpha-1} \quad (25)$$

We proved in *0.1: Limit Proof I* that $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$. Since 0 is constant and contains neither p nor q , $\lim_{q \rightarrow p} 0 = 0$. Therefore, by the squeeze theorem, $\lim_{q \rightarrow p} \left| \frac{f(p) - f(q)}{p - q} \right| = f'(p) = 0$ for all $p \in \mathbb{R}$.

Now, recall the mean value theorem: if f is a real, continuous function on $[a, b]$ which is differentiable on (a, b) , there is a point $x \in (a, b)$ where $f(b) - f(a) = (b - a)f'(x)$. Since our $f'(x) = 0$ for all $x \in \mathbb{R}$, fix p and let $q \in \mathbb{R}$. The Mean Value Theorem tells us that for the interval $[p, q]$, there exists a point $x \in \mathbb{R}$ such that $f(q) - f(p) = (q - p)f'(x) = (q - p) * 0 = 0$. Therefore, for any $q \in \mathbb{R}$, $f(p) = f(q)$. Ergo, f is constant. OEΔ

2 Problem 2

On the interval $(100, \infty)$, define functions $f(x) = x + \sin(x)$, $g(x) = x$, $p(x) = x + \sin(x) \cos(x)$, and $q(x) = p(x)e^{\sin(x)}$.

(a) Using facts we proved about limits and derivatives, prove that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1, \quad \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} \text{ DNE}, \quad \lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} \text{ DNE}, \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)} = 0.$$

(b) For both scenarios $\frac{f(x)}{g(x)}$ and $\frac{p(x)}{q(x)}$, explain why these results do not contradict L'Hôpital's rule.

Note: even though we haven't learned about sin and cos yet, you are free to use any familiar properties of them you wish.

2.1 Solution (a)

We prove that $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$.

Proof. First write out the limit.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x + \sin(x)}{x} = \lim_{x \rightarrow \infty} \left[1 + \frac{\sin(x)}{x} \right] \quad (26)$$

By limit laws, we can separated added up properties.

$$\lim_{x \rightarrow \infty} \left[1 + \frac{\sin(x)}{x} \right] = \lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} \quad (27)$$

Since 1 is constant, $\lim_{x \rightarrow \infty} 1 = 1$. My first assertion about sin: it is bounded to $[-1, 1]$ and is cyclical, so $-1 \leq \sin(x) \leq 1$. Divide all sides by x to find $-\frac{1}{x} \leq \frac{\sin(x)}{x} \leq \frac{1}{x}$. Take the limit as x approaches ∞ to get $-\lim_{x \rightarrow \infty} \frac{1}{x} = \lim_{x \rightarrow \infty} -\frac{1}{x} \leq \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} \leq \lim_{x \rightarrow \infty} \frac{1}{x}$. We proved earlier that $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$, and by extension, $-\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Therefore, by the sequence squeeze theorem, $\lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = 0$. Put this back into $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = 1 + 0 = 1 \quad (28)$$

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We prove that $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$ does not exist.

Proof. My second assertion is that if $f(x) = \sin(x)$, $f'(x) = \cos(x)$. Since $(a + b)' = a' + b'$, $(x + \sin(x))' = (x') + (\sin(x))' = 1 + \cos(x)$.

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} [1 + \cos(x)] \quad (29)$$

Create two sequences in terms of $n \in \mathbb{N}$ that diverge to ∞ as $n \rightarrow \infty$.

$$a_n = 2n\pi; b_n = (2n + 1)\pi \quad (30)$$

Apply both a_n and b_n to $1 + \cos(x)$.

$$1 + \cos(a_n) = 1 + \cos(2n\pi) = 1 + 1 = 2 \quad (31)$$

$$1 + \cos(b_n) = 1 + \cos((2n+1)\pi) = 1 - 1 = 0 \quad (32)$$

This gives us two subsequences that converge to two different values as $x \rightarrow \infty$. Therefore, the limit does not exist. OEΔ

We prove that $\lim_{x \rightarrow \infty} \frac{p(x)}{q(x)}$ does not exist.

Proof.

$$\lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} = \lim_{x \rightarrow \infty} \frac{p(x)}{p(x)e^{\sin(x)}} = \lim_{x \rightarrow \infty} e^{-\sin(x)} \quad (33)$$

Define two sequences as $n \rightarrow \infty$.

$$a_n = \left(2n + \frac{1}{2}\right)\pi; b_n = \left(2n + \frac{3}{2}\right)\pi$$

Apply both a_n and b_n to $e^{-\sin(x)}$.

$$e^{-\sin(a_n)} = e^{-\sin((2n+\frac{1}{2})\pi)} = e^{-1} \quad (34)$$

$$e^{-\sin(b_n)} = e^{-\sin((2n+\frac{3}{2})\pi)} = e^1 = e \quad (35)$$

This gives us two subsequences that converge to two different values as $x \rightarrow \infty$. Therefore, the limit does not exist. OEΔ

We prove that $\lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)} = 0$.

Proof. First, we find the derivative of $p(x)$ and $q(x)$. This involves the multiplication rule $((fg)' = fg' + f'g)$, the chain rule $((f(g))' = g'f'(g))$, my assertion that $(\cos(x))' = -\sin(x)$, and the $\frac{d}{dx}e^x = e^x$. I was given verbal approval in Office Hours to assert that $\frac{d}{dx}e^x = e^x$.

$$p'(x) = (x + \sin(x) \cos(x))' = 1 + (\sin(x) \cos(x))' = 1 + \cos^2(x) - \sin^2(x) \quad (36)$$

$$= 1 + \cos(2x) = 2 \cos^2(x) \quad (37)$$

$$q'(x) = \left(p(x)e^{\sin(x)}\right)' = p'(x)e^{\sin(x)} + p(x) \left(e^{\sin(x)}\right)' \quad (38)$$

$$= p'(x)e^{\sin(x)} + p(x) \cos(x)e^{\sin(x)} \quad (39)$$

$$= 2 \cos^2(x)e^{\sin(x)} + (x + \sin(x) \cos(x)) \cos(x)e^{\sin(x)} \quad (40)$$

$$= e^{\sin(x)} \cos(x) (2 \cos(x) + x + \sin(x) \cos(x)) \quad (41)$$

Now, we compute $\frac{p'(x)}{q'(x)}$.

$$\frac{p'(x)}{q'(x)} = \frac{2 \cos^2(x)}{e^{\sin(x)} \cos(x) (2 \cos(x) + x + \sin(x) \cos(x))} \quad (42)$$

$$= \frac{2 \cos(x)}{e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x))} \quad (43)$$

Take a few inequalities based on the bounds of $\sin(x)$ and $\cos(x)$ being $-1 \leq \sin(x) \leq 1$ and $-1 \leq \cos(x) \leq 1$.

$$e^{-1} \leq e^{\sin(x)} \leq e \quad (44)$$

$$-2 \leq 2 \cos(x) \leq 2 \quad (45)$$

$$-1 \leq \sin(x) \cos(x) \leq 1 \quad (46)$$

$$-3 \leq 2 \cos(x) + \sin(x) \cos(x) \leq 3 \quad (47)$$

$$x - 3 \leq 2 \cos(x) + x + \sin(x) \cos(x) \leq x + 3 \quad (48)$$

$$e^{-1} (x - 3) \leq e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x)) \leq e (x + 3) \quad (49)$$

$$\frac{1}{e^{-1} (x - 3)} \geq \frac{1}{e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x))} \geq \frac{1}{e (x + 3)} \quad (50)$$

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3 Problem 3

Let $f : (a, b) \rightarrow \mathbb{R}$ be continuous. Let $c \in (a, b)$, and suppose f is differentiable on (a, c) and (c, b) . Suppose that

$$\lim_{x \rightarrow c} f'(x)$$

exists and is a real number. Show that f is differentiable at c . *Hint:* Mean value theorem.

3.1 Solution

Proof. Let $\lim_{x \rightarrow c} f'(x) = L$. Recall the definition of $f'(c)$.

$$f'(c) = \lim_{t \rightarrow c} \frac{f(t) - f(c)}{t - c} \quad (51)$$

Without loss of generality, suppose that $t > c$. By the Mean Value Theorem, there exists $d \in (c, t)$ where $\frac{f(t) - f(c)}{t - c} = f'(d)$. Since $d \in (c, t)$, $c < d < t$, so $0 < d - c < t - c$ and therefore $0 < |d - c| < |t - c|$.

$f'(c) = \lim_{t \rightarrow c} \frac{f(t) - f(c)}{t - c}$ means that for all $\varepsilon > 0$, there exists $\delta > 0$ where $|t - c| < \delta$ implies $\left| \frac{f(t) - f(c)}{t - c} - f'(c) \right| < \varepsilon$. Since $|d - c| < |t - c|$, $|d - c| < \delta$. Furthermore, since $\frac{f(t) - f(c)}{t - c} = f'(d)$, $|f'(d) - f'(c)| < \varepsilon$. Therefore, for all $\varepsilon > 0$, there exists $\delta > 0$ where $|d - c| < \delta$ implies $|f'(d) - f'(c)| < \varepsilon$. This means $f'(c) = \lim_{d \rightarrow c} f'(d)$. It is given that $\lim_{x \rightarrow c} f'(x)$ exists and is a real number. Therefore $f'(c)$ exists and is a real number. Ergo, f is differentiable at c . $\square$

4 Problem 4

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be C^2 (meaning its second derivative exists and is continuous on $\mathbb{R}$).

- (a) Let $x \in \mathbb{R}$. For each $h > 0$, show there are numbers ξ and η with $x - h < \xi < x < \eta < x + h$ and

$$\frac{f''(\xi) + f''(\eta)}{2} = \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

- (b) Show that, for each $x \in \mathbb{R}$,

$$f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

- (c) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *convex* if for all $x, y \in \mathbb{R}$ and $0 \leq \lambda \leq 1$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Geometrically, this means the graph of f lies *below* all of its secant lines. Show that any C^2 convex function f satisfies $f''(x) \geq 0$ for all x . *Hint:* convexity tells you something about the numerator in part (b).

- (d) Conversely, if a twice-differentiable function f satisfies $f''(x) \geq 0$ for all x , show that f is convex. *Hint:* For $\lambda \in (0, 1)$ and $x < y \in \mathbb{R}$, the number $z := \lambda x + (1 - \lambda)y$ satisfies $x < z < y$. Apply the mean value theorem on $[x, z]$ and $[z, y]$ to get an expression for $\lambda f(x) + (1 - \lambda)f(y)$.

4.1 Solution (a)

Proof. Recall Taylor's Theorem, for some $\gamma \in (\alpha, \beta)$ or $\gamma \in (\beta, \alpha)$.

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k \quad (52)$$

$$f(\beta) = P(\beta) + \frac{f^{(n)}(\gamma)}{n!} (\beta - \alpha)^n \quad (53)$$

Let $n = 2$ and let $\alpha = x$ and $\beta = x + h$. Some $\eta \in (x, x + h)$ will satisfy Taylor's Theorem for this.

$$P(t) = f(x) + f'(x)(t - x) \quad (54)$$

$$f(x + h) = f(x) + f'(x)h + \frac{f''(\eta)h^2}{2} \quad (55)$$

Let $n = 2$ and let $\alpha = x$ and $\beta = x - h$. Some $\xi \in (x - h, x)$ will satisfy Taylor's Theorem for this.

$$P(t) = f(x) + f'(x)(t - x) \quad (56)$$

$$f(x - h) = f(x) - f'(x)h + \frac{f''(\xi)h^2}{2} \quad (57)$$

Add the latter two of each of these equalities together.

$$f(x + h) + f(x - h) = f(x) + f'(x)h + \frac{f''(\eta)h^2}{2} + f(x) - f'(x)h + \frac{f''(\xi)h^2}{2} \quad (58)$$

Rearrange this to isolate $\frac{(f''(\eta)+f''(\xi))h^2}{2}$, then divide both sides by h^2 .

$$f(x+h) + f(x-h) - f(x) - f(x) - f'(x)h + f'(x)h = \frac{f''(\eta)h^2}{2} + \frac{f''(\xi)h^2}{2} \quad (59)$$

$$f(x+h) + f(x-h) - 2f(x) = \frac{(f''(\eta) + f''(\xi))h^2}{2} \quad (60)$$

$$\frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = \frac{f''(\eta) + f''(\xi)}{2} \quad (61)$$

This is the equation we were looking for.

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4.2 Solution (b)

Proof. We know that for each $h > 0$, show there are numbers ξ and η with $x-h < \xi < x < \eta < x+h$ and

$$\frac{f''(\xi) + f''(\eta)}{2} = \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}. \quad (62)$$

$f : \mathbb{R} \rightarrow \mathbb{R}$ and $f'' : \mathbb{R} \rightarrow \mathbb{R}$ are both continuous. This means that for all $\varepsilon > 0$, there exists $\delta > 0$ where $0 < |x+h-x| = |h| < \delta$ (or $0 < |x-h-x| = |h| < \delta$) implies $|f''(x+h) - f''(x)| < \varepsilon$ (or $|f''(x-h) - f''(x)| < \varepsilon$). Since $x-h < \xi < x$, $x-h-x < \xi-x < 0$ and therefore $|\xi-x| < |h|$ and therefore $|f''(\xi) - f''(x)| < \varepsilon$. Similarly, we can say that for the η prior where $x < \eta < x+h$, $|f''(\eta) - f''(x)| < \varepsilon$. Add these together to find that $|f''(\xi) - f''(x)| + |f''(\eta) - f''(x)| < 2\varepsilon$. By the triangle inequality, $|f''(\xi) + f''(\eta) - 2f''(x)| < 2\varepsilon$. Divide both sides by 2 and find $\left| \frac{f''(\xi) + f''(\eta)}{2} - f''(x) \right| < \varepsilon$. Using the same value of h as before and using (62), we find that $|h| < \delta$ implies that $\left| \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} - f''(x) \right| < \varepsilon$.

Ergo, $\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x)$.

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5 Problem 5

Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Compute the upper and lower Riemann integrals of f . Is f Riemann integrable?

5.1 Solution

The issue here is the density of the rationals in the reals.

First we prove the upper Riemann integral.

$$\overline{\int_0^1} f(x) dx = 1 \quad (63)$$

Proof. Let P be any partition of $[0, 1]$ with $n + 1$ elements. Take any $j \in \mathbb{N}$ where $1 \leq j \leq n$. By the density of the rationals in the reals, there exists $x \in [x_{j-1}, x_j]$ where $x \in \mathbb{Q}$. Since the only two possible values of $f(x)$ are 0 and 1, $\sup_{x \in [x_{j-1}, x_j]} f(x) = 1$. We can compute the upper sum of f from here.

$$U(P, f) = \sum_{j=1}^n \Delta x_j \sup_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^n \Delta x_j * 1 = \sum_{j=1}^n \Delta x_j \quad (64)$$

Since the sum of Δx_j telescopes, $\sum_{j=1}^n \Delta x_j = x_n - x_0$. Therefore, for any partition P of $[0, 1]$, $U(P, f) = 1 - 0 = 1$. Since $U(P, f) = 1$ for every partition P of $[0, 1]$, $\inf\{U(P, f) | \text{Partition } P\} = 1$. Ergo, $\overline{\int_0^1} f(x) dx = \inf\{U(P, f) | \text{Partition } P\} = 1$. OEΔ

Now we prove the lower Riemann integral.

$$\underline{\int_0^1} f(x) dx = 0 \quad (65)$$

Proof. Let P be any partition of $[0, 1]$ with $n + 1$ elements. Take any $j \in \mathbb{N}$ where $1 \leq j \leq n$. By the density of the irrationals in the reals, there exists $t \in [x_{j-1}, x_j]$ where $t \notin \mathbb{Q}$. Since the only two possible values of $f(x)$ are 0 and 1, $\inf_{t \in [x_{j-1}, x_j]} f(t) = 0$. We can compute the upper sum of f from here.

$$L(P, f) = \sum_{j=1}^n \Delta x_j \inf_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^n \Delta x_j * 0 = \sum_{j=1}^n 0 = 0 \quad (66)$$

Therefore, for any partition P of $[0, 1]$, $L(P, f) = 0$. Since $L(P, f) = 0$ for every partition P of $[0, 1]$, $\sup\{L(P, f) | \text{Partition } P\} = 0$. Ergo, $\underline{\int_0^1} f(x) dx = \sup\{L(P, f) | \text{Partition } P\} = 0$. OEΔ

Since $\overline{\int_0^1} f(x) dx \neq \underline{\int_0^1} f(x) dx$, the function is not Riemann integrable on $[0, 1]$.